

If an $n \times n$ matrix has a repeated eigenvalue r with multiplicity m , what are the possible dimensions of the eigenspace corresponding to eigenvalue r ?

Answer:

If an $n \times n$ matrix has a repeated eigenvalue r with multiplicity m , the possible dimensions of the eigenspace corresponding to the eigenvalue r can range from 1 to m . Specifically, the dimension of the eigenspace E_r must satisfy:

$$1 \leq \dim E_r \leq m.$$

This means that the eigenspace dimension could be any integer from 1 to m . If the dimension of the eigenspace equals the multiplicity m of the eigenvalue m for every eigenvalue, this suggests that the matrix is diagonalizable. If the dimension is less m , then the matrix is not diagonalizable.